ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
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Project Name _____
Project Number _____

17.7CP Structural Checklist for Building Type S4: Dual Systems with Backup Steel Moment Frames and Stiff Diaphragms

Table 17-14. Collapse Prevention Structural Checklist for Building Type S4.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of braced frames or shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1 A.3.3.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DRIFT CHECK: The drift ratio of the steel moment frames acting alone, calculated using the Quick Check procedure of Section 4.4.3.1 using 25% of V_c, is less than 0.025.	5.5.2.1.2	A.3.1.3.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN AXIAL STRESS CHECK: The axial stress caused by gravity loads in frame columns subjected to overturning forces is less than $0.10F_y$. Alternatively, the axial stress caused by overturning forces alone, calculated using the Quick Check procedure of Section 4.4.3.6 is less than $0.30F_y$.	5.5.2.1.3	A.3.1.3.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BRACE AXIAL STRESS CHECK: The axial stress in the diagonal braces, calculated using the Quick Check procedure of Section 4.4.3.4 and neglecting the steel moment frame is less than $0.50F_y$.	5.5.4.1	A.3.3.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE BEARING WALLS: Floor and roof girders and trusses are not supported at the ends of concrete walls that are less than 10 in. (254 mm) thick. This statement only applies to framing supports located less than two times the wall thickness away from the wall end.	5.5.2.5.1	A.3.1.6.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the concrete shear walls, calculated using the Quick Check procedure of Section 4.4.3.3, and neglecting the steel moment frame, is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f'_c}$.	5.5.3.1.1	A.3.2.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REINFORCING STEEL: The ratio of reinforcing steel area to gross concrete area is not less than 0.0012 in the vertical direction and 0.0020 in the horizontal direction.	5.5.3.1.3	A.3.2.2.2	

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Connections						
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad.	5.7.3.1	A.5.3.1
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of seismic forces to the shear walls.	5.7.2	A.5.2.1
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO STEEL FRAMES: Diaphragms are connected for transfer of seismic forces to the steel frames.	5.7.2	A.5.2.2
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FOUNDATION DOWELS: Wall reinforcement is doweled into the foundation.	5.7.3.4	A.5.3.5

Status				**Evaluation Statement**	**Tier 2 Reference**	**Commentary Reference**	**Comments**
Moderate Seismicity (Complete the Following Items in Addition to the Items for Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: For braced frames, the number of braced bays in each line is greater than 2.	5.5.1.1	A.3.3.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the strength of the adjoining members based on the specified minimum yield stress of the steel.	5.5.2.2.1	A.3.1.3.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COMPACT MEMBERS: All moment frame and brace elements meet section requirements in accordance with AISC 360, Table B4.1a.	5.5.2.2.5	A.3.1.3.8	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONNECTION STRENGTH: All the brace connections develop the buckling capacity of the diagonals.	5.5.4.4	A.3.3.1.5	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	K-BRACING: The bracing system does not include K-braced bays.	5.5.4.6	A.3.3.2.1	

Status				**Evaluation Statement**	**Tier 2 Reference**	**Commentary Reference**	**Comments**
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the strength of the adjoining members or panel zones based on 110% of the expected yield stress of the steel per AISC 341, Section A3.2.	5.5.2.2.1	A.3.1.3.4	

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C	NC	N/A	U			
☐	☐	☐	☐	COLUMN SPLICES: At all column splice details located in moment frames, the web and flanges of I-shaped members, or all walls of tube/box members are connected to each other with the partial penetration welds with effective throat of at least 85% of the smaller member thickness or with plates that have been bolted or welded to the columns capable of developing $A_g F_{ye}$ of the thinner flange, web or tube/box wall.	5.5.2.2.4	A.3.1.3.6
C	**NC**	**N/A**	**U**	STRONG COLUMN—WEAK BEAM: The percentage of strong column–weak beam joints in each story of each line of moment frames is greater than 50%.	5.5.2.1.5	A.3.1.3.7
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	COMPACT MEMBERS: All moment-frame and brace elements meet section requirements in accordance with AISC 341, Table D1.1a, for moderately ductile members.	5.5.2.2.5 5.5.4.3	A.3.1.3.8
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	COLUMN SPLICES: All column splice details located in braced frames develop 50% of the tensile strength of the column.	5.5.4.2	A.3.3.1.3
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	SLENDERNESS OF DIAGONALS: All diagonal elements required to carry compression have *Kl/r* ratios less than 200.	5.5.4.3	A.3.3.1.4
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	CONNECTION STRENGTH: All the brace connections develop the yield capacity of the diagonals.	5.5.4.4	A.3.3.1.5
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	CHEVRON BRACING: Beams in chevron, or V-braced, bays are capable of resisting the vertical load resulting from the simultaneous yielding and buckling of the brace pairs.	5.5.4.6	A.3.3.2.3
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	CONCENTRICALLY BRACED FRAME JOINTS: All the diagonal braces frame into the beam–column joints concentrically.	5.5.4.8	A.3.3.2.4
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	COUPLING BEAMS: Coupling beams have stirrups spaced at or less than d/2, and each wall or wall segment connected to the coupling beam is supported such that it can resist shear and overturning forces in the absence of the coupling beam. This statement only applies to coupling beams with span-to-depth ratios exceeding 2-to-1.	5.5.3.2.1	A.3.2.2.3
☐	☐	☐	☐			

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Flexible Diaphragms						
C	**NC**	**N/A**	**U**	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	OPENINGS AT SHEAR WALLS: Diaphragm openings immediately adjacent to the shear walls are less than 15% of the wall length.	5.6.1.3	A.4.1.4
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	OPENINGS AT FRAMES: Diaphragm openings immediately adjacent to the braced frames or moment frames extend less than 25% of the frame length.	5.6.1.3	A.4.1.5
☐	☐	☐	☐			

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